Hemoglobin (Hb) A_{1c}

Test Details

METHODOLOGY

Roche Tina Quant

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

This test is useful for evaluating the long-term control of blood glucose concentrations in patients with diabetes, diagnosing diabetes and identifying patients at increased risk for diabetes (prediabetes).

Limitations

For diagnostic purposes, HbA1c values should be used in conjunction with information from other diagnostic procedures and clinical evaluations.[1]

This test is not intended to replace daily home testing of urine or blood glucose.[1]

Any cause of shortened erythrocyte survival will reduce exposure of erythrocytes to glucose with a consequent decrease in HbA1c.[1,2] Causes of shortened erythrocyte lifetime might be hemolytic anemia or other hemolytic diseases, homozygous sickle cell trait, pregnancy, or recent significant or chronic blood loss. Caution should be used when interpreting the HbA1c results from patients with these conditions.

American Diabetes Association (ADA) guidelines describe potential limitations in HbA1c measurements due to hemoglobin variants, assay interferences, ethnicity, age and conditions associated with altered red blood cell (RBC) turnover, all of which may necessitate the use of an alternative form of diagnostic glucose testing.[2]

Care must be taken when interpreting any HbA1c result from patients with Hemoglobin variants.[1-3] Abnormal hemoglobins might affect the half-life of the red cells or the in vivo glycation rates. In these cases, even analytically correct results do not reflect the same level of glycemic control that would be expected in patients with normal hemoglobin.[2] In the case of assay interference, marked discordance between measured HbA1c and observed plasma glucose concentrations should prompt an investigation into the presence of hemoglobin variants that may interfere with test results.[1,2]

Glycated hemoglobin F (HbF) is not detected as it does not contain the glycated β-chain that characterizes HbA1c.[1] Specimens containing high amounts of HbF (>10 %) may result in lower than expected HbA1c values.[1]

Custom Additional Information

Hemoglobin A1c results from the non-enzymatic glycation of the amino (N)-terminal valine residue of hemoglobin A. This process is dependent on average glucose concentrations and occurs throughout the 120-day lifespan of the RBC. Therefore, HbA1c reflects glycemic control over the previous three months.[4-6] HbA1c represents a weighted average, with approximately 50% of the value due to the mean blood glucose (BG) concentrations in the 30 days prior to sampling; BG concentrations from the previous 90 to 120 days make up about 10% of the final total HbA1c value.[5]

The ADAG (A1c-Derived Average Glucose) study found a strong correlation between the HbA1c and estimated average glucose concentrations.[7]A change (either positive or negative) in HbA1c percentage of 0.5% is considered clinically significant.[8,9]

The American Diabetes Association's Standards of Medical Care in Diabetes, published in 2018, addressed the utilization of HbA1c in the diagnosis and management of Diabetes Mellitus (DM).[1] The diagnosis of DM is made when the HbA1c values are >6.5% based on an NGSP-certified test. Prediabetes is defined by an HbA1c of 5.7% to 6.4%. Patients with prediabetes should be tested yearly in order to determine whether they have converted to diabetic status. Plasma glucose concentrations are recommended over HbA1c testing for diagnosing Type 1 DM patients who have overt symptoms of hyperglycemia, most of whom are pediatric patients. HbA1c, fasting plasma glucose and 2-hour plasma glucose values obtained during oral glucose tolerance testing are equally beneficial in diagnosing Type 2 DM in both younger and older patients.[1]

The ADA guidelines recommend that HbA1c testing be performed at least twice yearly in diabetic patients who have achieved stable glycemic control.[2] For those patients who are not at goal or for whom therapy recently changed, quarterly HbA1c testing is recommended. The guidelines also caution that HbA1c does not measure glycemic variability or hypoglycemic risk, although hypoglycemia is less common among patients with HbA1c values of <7.0% to 7.5%.[2] The ADA offers guidelines for initiating and escalating therapy based on HbA1c concentrations.

Specimen Requirements

Specimen

Whole blood

Volume

4 mL

Minimum Volume

0.5 mL (**Note**: This volume does **not** allow for repeat testing.)

Pediatric EDTA whole blood tubes may be used. Please place original labeled capillary tube in a labeled transport tube for shipment to the laboratory.

Container

Lavender-top (EDTA) tube, green-top (lithium heparin) tube, or gray-top (sodium fluoride) tube

Collection Instructions

The usual precautions in the collection of venipuncture samples should be observed. The sample must be free of clots. Samples with any hematocrit disorders can lead to erroneous results. Send the entire tube to the laboratory.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

- Hemoglobin (Hb) A_{1c}: 4.8% to 5.6%

- Prediabetes: 5.7% to 6.4%

- Diabetes: ≥6.5%

- Glycemic control for adults with diabetes: <7.0%

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Clotted specimen

Footnotes

1. Tina Quant Hemoglobin A1c Gen.3 - Hemolysate application [package insert]. Roche Diagnostics, Mannheim Germany. Version 1.0: 2017-06.
2. American Diabetes Association. 2. Classification and diagnosis of diabetes: standards of medical Care in diabetes-2021. *Diabetes Care.* 2021 Jan;44(Suppl 1):S15-S33. PubMed 33298413 (http://www.ncbi.nlm.nih.gov/pubmed/33298413)

3. Little RR, La'ulu SL, Hanson SE, Rohlfing CL, Schmidt RL. Effects of 49 different rare Hb variants on HbA1c measurement in eight methods. *J Diabetes Sci Technol.* 2015 Jul;9(4):849-856. PubMed 25691657 (http://www.ncbi.nlm.nih.gov/pubmed/25691657)

4. Sazudek CD, Brick JC. The clinical use of hemoglobin A1c. *J Diabetes Sci Technol.* 2009 Jul;3(4):629-634. PubMed 20144304 (http://www.ncbi.nlm.nih.gov/pubmed/20144304)

5. Hare MJL, Shaw Je, Zimmet PZ. Current controversies in the use of hemoglobinA1c. *J Intern Med.* 2012 Mar;271(3):227-236. PubMed 22333004 (http://www.ncbi.nlm.nih.gov/pubmed/22333004)

6. Rhea JM, Molinaro R. Pathology consultation on HbA1c methods and interferences. *Am J Clin Pathol.* 2014 Jan;141(1):5-16. PubMed 24343732 (http://www.ncbi.nlm.nih.gov/pubmed/24343732)

7. Nathan DM, Kuenen J, Borg R, et al. Translating the A1c assay into estimated average glucose values. *Diabetes Care.* 2008 Aug;31(8):1473-1478. PubMed 18540046 (http://www.ncbi.nlm.nih.gov/pubmed/18540046)

8. American Diabetes Association. 6. Glycemic targets: standards of medical care in diabetes-2021. *Diabetes Care.* 2021 Jan;44(Suppl 1):S73-S84. PubMed 33298417 (http://www.ncbi.nlm.nih.gov/pubmed/33298417)

9. Little RR, Rohlfing CL. The long and winding road to optimal HbA1c measurement. *Clin Chim Acta.* 2013 Mar 15;418:63-71. PubMed 23318564 (http://www.ncbi.nlm.nih.gov/pubmed/23318564)